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PAPER

Design and stable flight of a 21 g insect-like tailless flapping wing micro air vehicle with angular rates feedback control

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Abstract

An insect-like tailless flapping wing micro air vehicle (FW-MAV) without feedback control eventually becomes unstable after takeoff. Flying an insect-like tailless FW-MAV is more challenging than flying a bird-like tailed FW-MAV, due to the difference in control principles. This work introduces the design and controlled flight of an insect-like tailless FW-MAV, named KUBeetle. A combination of four-bar linkage and pulley-string mechanisms was used to develop a lightweight flapping mechanism that could achieve a high flapping amplitude of approximately 190°. Clap-and-flings at dorsal and ventral stroke reversals were implemented to enhance vertical force. In the absence of a control surface at the tail, adjustment of the location of the trailing edges at the wing roots to modulate the rotational angle of the wings was used to generate control moments for the attitude control. Measurements by a 6-axis load cell showed that the control mechanism produced reasonable pitch, roll and yaw moments according to the corresponding control inputs. The control mechanism was integrated with three sub-micro servos to realize the pitch, roll and yaw controls. A simple PD feedback controller was implemented for flight stability with an onboard microcontroller and a gyroscope that sensed the pitch, roll and yaw rates. Several flight tests demonstrated that the tailless KUBeetle could successfully perform a vertical climb, then hover and loiter within a 0.3 m ground radius with small variations in pitch and roll body angles.

1. Introduction

While both birds and insects flap their wings to produce flight forces, their flight principles are essentially different [1]. In most birds, except for the humming bird, the flapping stroke plane is nearly vertical during flight. Additionally, the useful aerodynamic forces are largely produced from downstroke motion [1, 2]. Therefore, most birds cannot hover. Compared to birds, insects flap their wings faster, and with greater amplitude, and produce forces during both downstrokes and upstrokes in a nearly horizontal plane [3]. Therefore, insects can perform remarkable hovering in confined spaces at low speeds.

The principles of attitude control in birds and insects are also quite different. Birds use their tail wing as a control surface to generate control moments for attitude change. On the other hand, without a tail control surface, insects rely on flapping wings to synchronously produce flight forces

and control moments. Insects are able to actively modify wing kinematics by changing the flapping amplitude, angle of attack [3, 4] and flapping frequency [5, 6] during flapping flight to produce control moments, as results of asymmetric aerodynamic forces in the two wings. These features can be observed in many insects such as fruit flies, locusts, dragonflies, hoverflies and hawkmoth. In more detail, Diptera *Drosophila melanogaster* [7] and hoverflies *Eristalis tenax* and *Episyrphus balteatus* [8] were found to shift the flapping angle ranges of the two wings simultaneously with respect to the mid-stroke while maintaining the flapping angle amplitude for pitch control. Kinematics studies on tethered locusts indicated that locusts modify their AoA of the forewing during pronation to resist nose-up disturbance [9–11]. For lateral control including roll and yaw responses, modification of flapping amplitudes in the two wings can be found in *Drosophila* spp. [12, 13], *Musca domestica* [14], *Calliphora* spp. [15, 16], dragonflies [17], beetles [18, 19], moths [20] and bugs [21]. In addition to

the wing kinematics modulations, some insects were also found to modify the location of the center of gravity (CG) by the change in postures for attitude control and stability augmentation [22, 23].

Therefore, reproducing such a complex ability in the form of a flapping wing micro air vehicle (FW-MAV) needs a generation of engineering design, where flapping wing kinematics can be modified to produce control moments as well as a flight force large enough to overcome body weight at the same time. Up until now, only a few research teams could successfully demonstrate the controlled flight of an insect-like tailless two-winged FW-MAV. With a weight of less than 20 g, the Nano Hummingbird is the first tailless FW-MAV that could successfully fly with only two flapping wings [24]. The Nano Hummingbird uses pulley-string mechanisms to create flapping wing motions and to produce control moments. This vehicle is mechanically more sophisticated than any other FW-MAV ever built, demonstrating extraordinary flight performance. In the sense that it uses only flapping wings for controlled flight, it actually mimics insect flight rather than the flight of a hummingbird. The Robobee is the smallest insect-like FW-MAV with a weight of only 80 mg [25–28]. However, its flight was performed with an off-board power supply and electronic control systems. Additionally, a heavier version of a two-winged, hummingbird-like FW-MAV (62 g) was also developed and demonstrated a short hovering of about 6 s [29].

We have been studying and mimicking the flight of a Rhinoceros beetle, which is capable of hovering flight [30–41]. In this paper, we present the design and upright stable flight of a 21 g insect-like tailless FW-MAV, named KUBeetle. Our flapping mechanism, which was first introduced in the previous works [39, 41], combines a four-bar linkage and a pulley-string mechanism designed to convert the rotary motion of a coreless DC motor to the flapping motion of the wings. This combination aims to enlarge the flapping amplitude to the level that limits the use of linkage mechanism only. Note that the 19 g Nano Hummingbird in [24] used string-based flapping mechanism entirely. Clap-and-fling mechanisms at the dorsal and ventral stroke reversals were thereby implemented for vertical force enhancement because of the large flapping motion [41]. A twisted wing (similar to [24]) for high flight performance found in our previous works [39, 40] was also used in the KUBeetle to mimic flapping wing feature of a real beetle. For attitude control, the KUBeetle was equipped with wing rotational modulation mechanisms, which require less actuation torque from actuators in our FW-MAV, that can independently modify each wing's rotation to produce pitch, roll, and yaw moments. Although a similar wing rotational modulation concept can be found in [24]; however, the control moment generators are based on our new and

simple linkage design operated by three conventional micro servos. Due to the instability of the FW-MAV, a proportional-derivative (PD) close-loop controller was then used to improve the KUBeetle's upright stability with a 3-axis MEMS gyroscope. Several flight tests were finally conducted and captured by three synchronized high-speed cameras to verify the flight stability of the KUBeetle. Thus, technical difficulties with lightweight parts, torque generation mechanism design, active control implementation, and compact onboard system are hurdles to overcome in this work for demonstrating controlled flight. While the Nano Hummingbird [24] is lack of its detailed information, this work may provide useful information to those, who are interested in developing a tailless FW-MAV that capable of hovering flight.

2. Flapping-wing mechanism

2.1. Design and fabrication of the flapping mechanism

The Rhinoceros beetle has the ability to flap its hind wing with a high amplitude (about 180°), at frequencies ranging from 30 to 40 Hz during a hover [32, 38]. To recreate this high flapping amplitude, a flapping mechanism based on the combination of a four-bar linkage and a pulley-string mechanism was designed [39, 41]. Figure 1(a) shows a schematic of the four-bar linkage and pulley-string mechanism. Rotary motion of the crank (O_0O_1) fixed to the last reduction gear is converted to reciprocating motion for the large pulley, which is fixed to the rocker (O_2O_3), through a coupler (O_1O_2). The lengths of the linkages (O_0O_3 and O_2O_3) and the amplitude of the angle β are predetermined as the input parameters. Figure 1(b) shows that when the angle β approaches the maximum ($\beta_{\max}$) and minimum ($\beta_{\min}$) values, the joints O_0 , O_1 , and O_2 collinearly correspond to the maximum and minimum distances between the joints O_0 and O_2 , respectively. For the symmetric flapping motion, $\beta_{\min} = \beta_{\max}$.

Therefore, the length of the crank (O_0O_1) and the coupler (O_1O_2) can be determined as follows:

$$O_0O_1 = \frac{1}{2}(X - Y), \quad (1)$$

$$O_1O_2 = \frac{1}{2}(X + Y), \quad (2)$$

where, $X = \sqrt{O_0O_3^2 + O_2O_3^2 + 2O_0O_3 \cdot O_2O_3 \sin \beta_{\max}}$

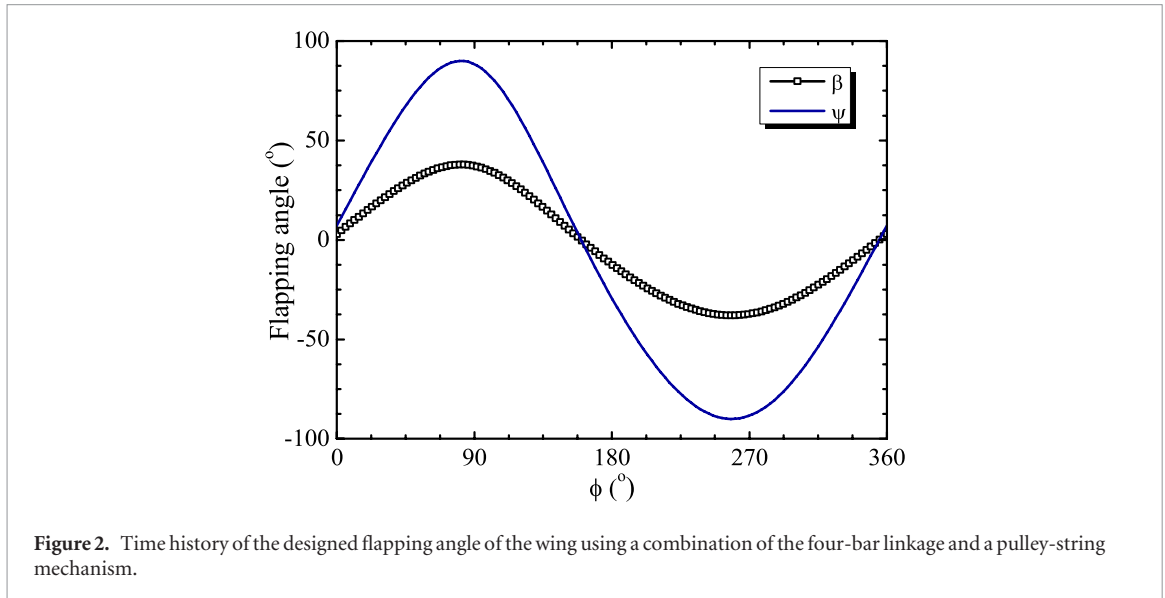
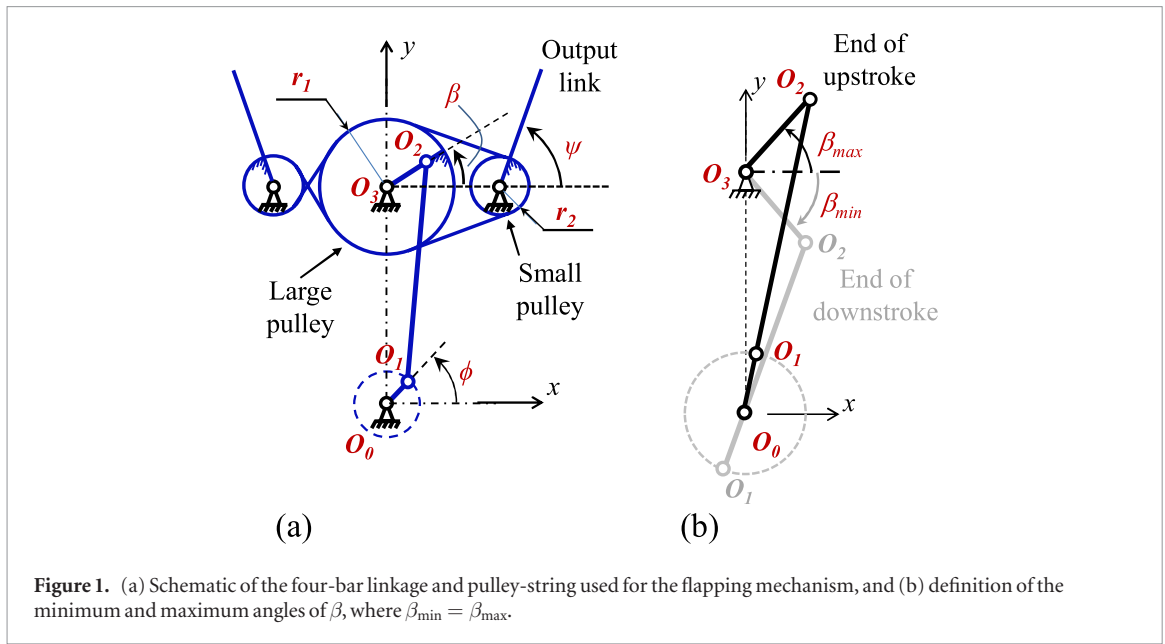
and $Y = \sqrt{O_0O_3^2 + O_2O_3^2 - 2O_0O_3 \cdot O_2O_3 \sin \beta_{\min}}$.

The angle β is then determined as the following equation:

$$\beta = -\sin^{-1}\left(\frac{Z}{W}\right) - \gamma, \quad (3)$$

where,

$$Z = \frac{O_0O_1^2 + O_2O_3^2 + O_0O_3^2 - O_1O_2^2 - 2O_0O_1 \cdot O_0O_3 \sin \phi}{2O_2O_3}, \quad (4)$$



$$W = \sqrt{(O_0O_1 \cos \phi)^2 + (O_0O_1 \sin \phi - O_0O_3)^2}, \quad (5)$$

$$\gamma = \tan^{-1} \left(\frac{O_0O_1 \cos \phi}{O_0O_1 \sin \phi - O_0O_3} \right). \quad (6)$$

The amplitude of the reciprocating motion of the large pulley (β) was amplified to the larger flapping motions of the two smaller pulleys (ψ) with strings. Note that the string between the large pulley and small pulley, shown on the left side in figure 1(a), was twisted to create the flapping motions of the two small pulleys in the same direction. One end of each wing's leading edge was connected to the output link. These were fixed to the small pulleys to create the wing flapping motion. The flapping angles ψ of the output link can be determined using the following equation:

$$\psi = \frac{r_1}{r_2} \beta, \quad (7)$$

where, r_1 and r_2 are the radii of the large and small pulleys, respectively, as figure 1(a) shows.

In order to reduce the disc loading for higher efficiency, the design flapping amplitude was set to 180° . This is similar to the beetle's hind wing during hover [32, 38]. Figure 2 shows that when the crank (O_0O_1) rotates to complete a circle, the output link sweeps the wing in one flapping cycle.

Figure 3(a) shows the 3D CAD design, from which all linkages and frames of the flapping mechanism were made from light-weight carbon/epoxy panels of 0.8 mm thickness. These were fabricated by CNC machine (MM-300S, resolution 10 μm , MANIX, Korea). Figure 3(b) shows the flapping mechanism that was manually assembled, which was of approximately 2.8 g weight. A 3.5 g coreless DC motor (Model CL720E2, China) was selected to drive the flapping wings. To effectively amplify the torque from the rotary motion of the motor, a number of gear sets with

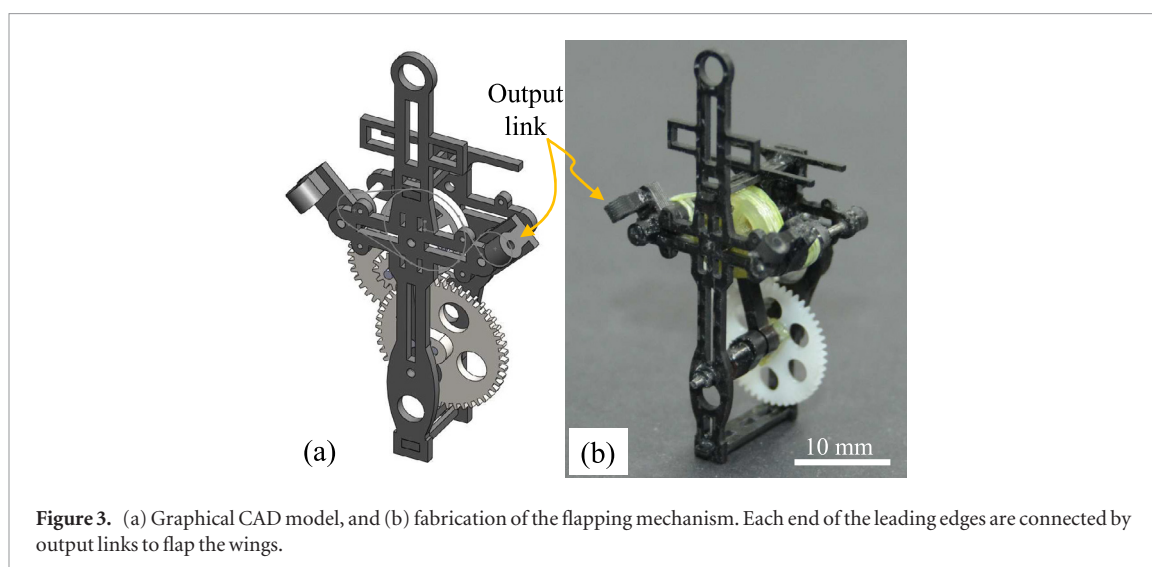


Figure 3. (a) Graphical CAD model, and (b) fabrication of the flapping mechanism. Each end of the leading edges are connected by output links to flap the wings.

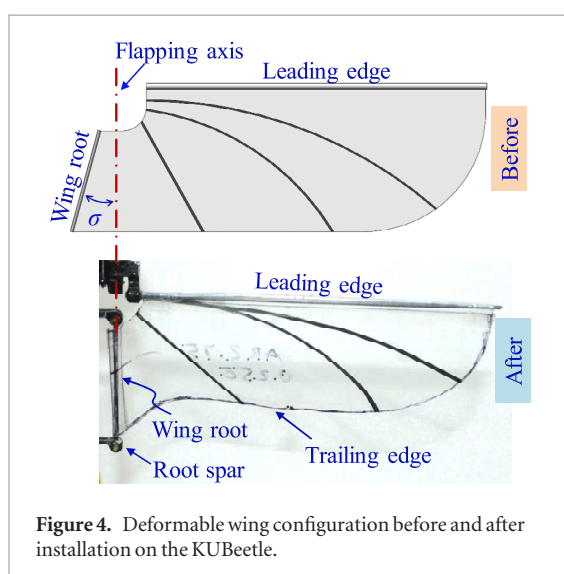


Figure 4. Deformable wing configuration before and after installation on the KUBeetle.

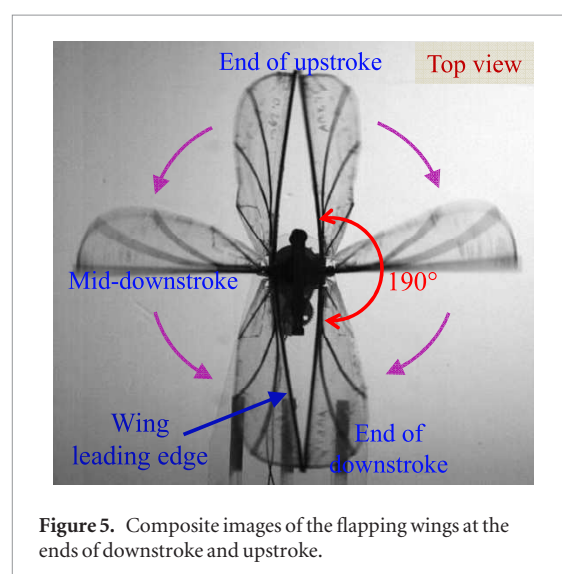


Figure 5. Composite images of the flapping wings at the ends of downstroke and upstroke.

different ratios were tested to obtain a high flapping frequency. The gear ratio of 20.6:1 was eventually found to produce the highest frequency, and was selected for the flapping mechanism.

2.2. Wing configuration for optimal efficiency

Using the unsteady blade element theory (UBET), a series of parametric studies were conducted to find the proper wing configuration for economic flight [39, 40]. The results revealed that to optimize the flight performance of the FW-MAV, a negatively twisted wing operating at a geometric angle of attack (AoA) approximating 40° was better than a flat wing. Our previous study also showed that the flapping wing, operating at low geometric AoA (about 40°) and high frequency with maximum force³/power² ratio, consumed about 26% less power than the wing, operating at high geometric AoA (about 53°) for maximum vertical force with lower frequency, to produce the same vertical force [40]. In addition, many insect wings perform a chordwise camber (which was not considered in [39, 40]), along with a spanwise twist during the flapping motion [35, 42–46]. This chordwise

camber, whose effect on the aerodynamic performance has been studied extensively [35, 43, 47], enhances force generation in the flapping wing. Therefore, the deformable wing configuration (twist and camber), which mimicked the beetle's hind wing, operating at a geometric AoA of about 40° at the mid-span, was selected for our KUBeetle.

The wings were made of light materials to reduce the effect of inertial force during the flapping motion. The wing leading edge made of carbon rod has a diameter of 1.2 mm and a length of 70 mm. Figure 4 shows that polyethylene terephthalate (PET) was used for the wing membrane. Margins of wing membrane along the leading edge and wing root were designed as sleeves to rotate freely about the carbon rod and the root spar, respectively [24]. The figure also shows that the trailing edge of the wing membrane at the root was extended to make the angle of $\sigma = 15^\circ$ with respect to the flapping axis. Hence, the root of the wing membrane was inserted on the root spar of the FW-MAV along the flapping axis, to create wing deformation during the flapping motion. Stiffeners made from one layer of carbon prepreg were placed on the wing membrane for

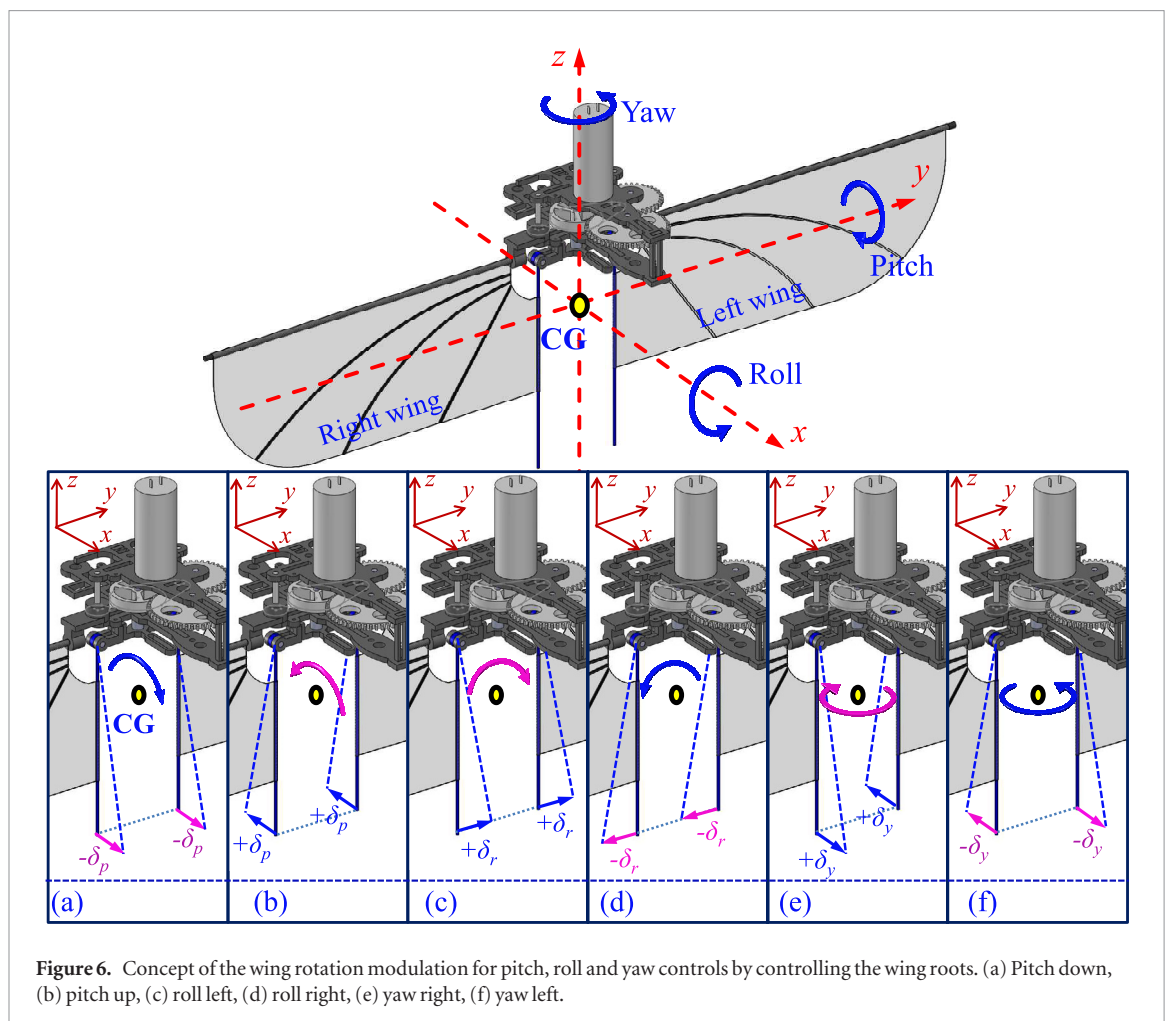


Figure 6. Concept of the wing rotation modulation for pitch, roll and yaw controls by controlling the wing roots. (a) Pitch down, (b) pitch up, (c) roll left, (d) roll right, (e) yaw right, (f) yaw left.

reinforcement, like the vein structures in the beetle's hind wing. The weight of each wing membrane with vein structures was approximately 0.12 g.

2.3. Clap-and-fling mechanism for thrust enhancement

Clap-and-fling, which was first explored by Weis-Fogh [48], occurs when the wings approach each other at the end of an upstroke (clap), and start the next stroke by separating the wings (fling) from leading to trailing edges [49]. Many insects utilize the clap-and-fling effect during flapping motion to enhance lift generation [48, 50–53]. The clap-and-fling effect has also been used in several FW-MAVs [54–56]. However, all these FW-MAVs used a four-winged mechanism to implement the clap-and-fling effect. Therefore, the flapping amplitude is relatively smaller than that of a hovering insect.

In an effort to mimic the flapping motion of a hovering Rhinoceros beetle (flapping its hind wings at a very high amplitude of 180° [32, 35, 38, 57]), our two-winged KUBeetle was designed to flap its wings to approximately 190°, resulting in the implementation of clap-and-fling mechanisms at the dorsal and ventral stroke reversals. Figure 5 shows that the tips of the left and right wings almost touch each other at the ends of the upstroke and downstroke. Measurement using a load cell showed that the clap-and-fling effect at both

stroke reversals improved the vertical force produced by the KUBeetle by approximately 16.2% [41]. This force improvement allows the KUBeetle to generate enough vertical force to lift its own weight.

3. Attitude control mechanisms

3.1. Concept of wing rotation modulation

For attitude change of the tailless KUBeetle, the wing rotation angle or geometric AoA is modified by synchronously tilting the locations of the wing roots at the trailing edges to generate control moments [37]. Figure 6 shows that to produce the pitch moment about the y-axis for forward or backward flight, the wing roots of the two wings are rotated about their hinges on the xz-plane in the same direction. Figure 6(a) shows that for forward flight, rotation of the wing roots of the left and right wings to a negative angle of δ_p generates a pitch-down moment. Figure 6(b) shows that for backward flight, rotation of the wing roots in both wings to a positive angle of δ_p produces a pitch-up moment. Roll moment around the x-axis, produced by asymmetric vertical forces in the left and right wings, creates lateral motion to the left or right. Figure 6(c) shows that to fly the FW-MAV to the left, rotation of the wing roots in both wings to the left on the yz-plane in a counter-clockwise direction for a positive angle of δ_r generates a roll left moment. Figure 6(d) shows that for a turn to

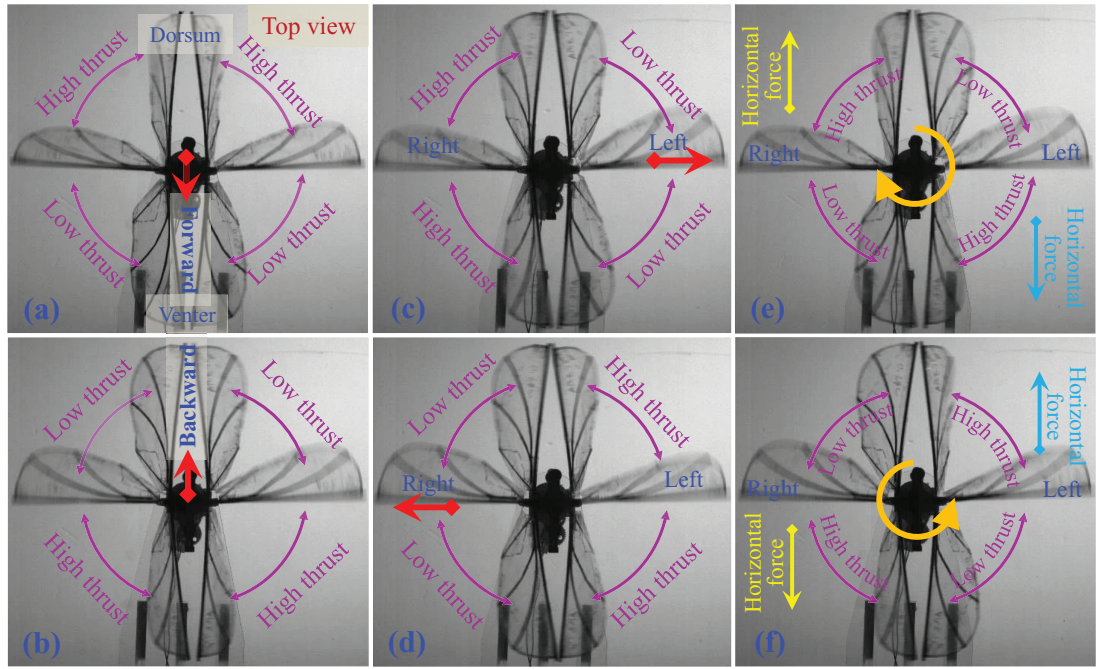


Figure 7. Composite images of the flapping wings captured by a high-speed camera from the top view of the FW-MAV. (a) Negative deflection angle of δ_p produces the pitch-down moment for forward flight. (b) Positive deflection angle of δ_p produces the pitch-up moment for backward flight. (c) Right wing produces higher thrust or vertical force than the left wing, resulting in generation of the roll-left moment to move the FW-MAV to the left. (d) Asymmetric thrust between the left and right wings generates roll-right moment to move the FW-MAV to the right. (e) Rotating the wing roots for a positive angle of δ_y in the opposite direction creates yaw-right moment. (f) Negative angle of δ_y results in the generation of yaw-left moment.

the right, rotating the wing roots to a negative angle of δ_r generates a right hand roll moment. Figures 6(e) and (f) show that synchronously rotating the wing roots of the two wings in the opposite direction on the xz -plane for positive and negative angles of δ_y produces yaw moments about the z -axis, resulting in changes to the heading direction to the right, and the left, respectively.

Figure 7 shows the composite images of the flapping wings captured by a high-speed camera at 2000 fps from the top view during the flapping motion. Figure 7(a) shows that tilting the wing roots for negative angle of δ_p on the xz -plane (figure 6(a)), results in the wings producing a high thrust in the dorsum, and a low thrust in the venter of the FW-MAV, resulting in the pitch-down moment for forward flight. On the other hand, when the wing roots rotate for a positive angle of δ_p , low thrust in the dorsum and high thrust in the venter of the FW-MAV produce a pitch-up moment for backward flight, as shown in figure 7(b). Figure 7(c) shows that high thrust in the right wing and low thrust in the left wing produces roll-left moment, resulting in lateral motion to the left. Figure 7(d) shows that the right wing producing lower thrust than the left wing produces a roll-right moment, resulting in the FW-MAV moving to the right. Figures 7(e) and (f) describe wing shapes for yaw-right and -left motions, respectively. Yaw moments can be created by rotating the wing roots of the left and right wings in the opposite direction, as shown in figures 6(e) and (f). Figure 7(e) shows that tilting the wing roots for a positive angle of δ_y produces horizontal force in the right wing, and directs it to the dorsum side; while that

of the left wing directs it to the venter side, resulting in generating a yaw-right moment. Figure 7(f) shows that a reversed control input with $-\delta_y$ in figure 6(f) can generate yaw-left moment.

3.2. Wing kinematics

Three synchronized high-speed cameras (Photron Ultima APX, Japan) were used to capture wing kinematics during the flapping motion while tilting the wing roots for attitude control. More details on the measurement of wing kinematics can be found in [33, 35, 39]. The flapping wing motion was tracked using the direct linear transformation (DLT) method to determine flapping angle and wing rotation angles along the wingspan (R) [58]. The raw data of these angles were fitted using the first six terms of the sine and cosine function, as follows:

$$\kappa(t) = \sum_{k=0}^6 [a_k \cos(2k\pi ft) + b_k \sin(2k\pi ft)], \quad (8)$$

where $\kappa(t)$ is either the flapping angle or the rotational angle, f is the flapping frequency, and a_k, b_k ($k = 0, 1, 2, \dots, 6$) are the coefficients according to the least-squares method.

Since the wings were designed to flap symmetrically about the xz -plane, only the wing kinematics of the right wing was captured. The deflection angle to produce pitch, roll, and yaw control moments was set to $\pm 6^\circ$. Figure 8 shows the time histories of the flapping angles and wing rotation angles (θ_r) in one flapping cycle. Figure 8(a) shows that the change of control

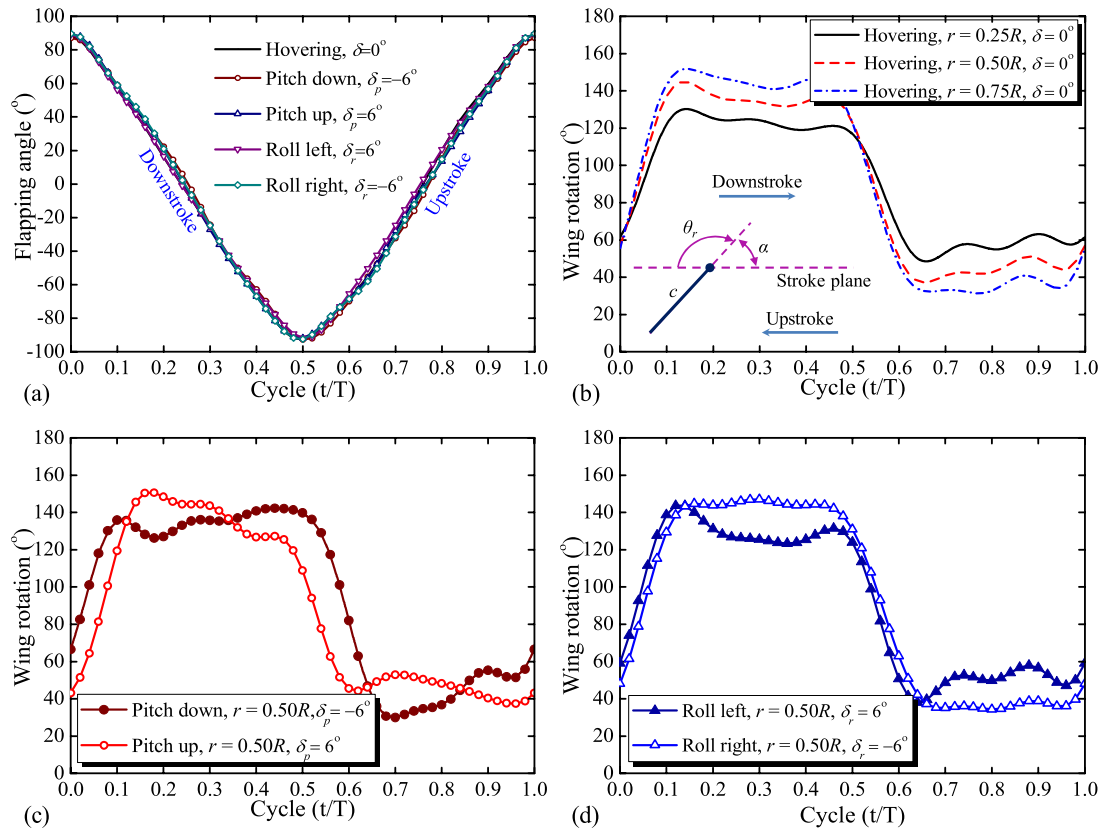


Figure 8. Wing kinematics of the right wing while tilting the deflection angle, $\delta = 6^\circ$, for attitude change: (a) flapping angles for different deflection of the wing root, (b) wing rotation angle for hovering or vertical flight, (c) rotation angles for pitch up and pitch down controls, and (d) wing rotation angles for roll-left and roll-right controls.

inputs did not significantly affect the time histories of the flapping angles. Figure 8(b) plots the time histories of the wing rotation angles at the three wing sections ($r/R = 0.25, 0.50$ and 0.75 , R is the wingspan) when the wing roots were vertically aligned to the flapping axes ($\delta_p = \delta_r = \delta_y = 0$), which is for vertical or hovering flight. Figure 8(c) plots the wing rotation angles at the wing section of $r/R = 0.50$ to produce pitch-down and -up moments. Figure 8(d) plots the time histories of the wing rotation angles for the right wing for roll control ($r/R = 0.50$). Figure 8(d) shows that the roll-left control causes relatively low wing rotation angles (θ_r in figure 8(b)), or high geometric AoA (α) for the right wing. This results in high thrust for the right wing, and a roll-left moment. On the other hand, the roll-right control generates high wing rotation angles (or low geometric AoA) (figure 7(d)), which causes low thrust in the right wing, generating a roll-right moment. For a yaw-right motion, the wing rotation angles to produce pitch-down moment (as shown in figure 8(c)) are assigned to the right wing, while the wing rotation angles for a pitch-up moment generation are used in the left wing. Reversing the control inputs for a yaw-right motion in the left and right wings generates the yaw-left moment.

3.3. Measurement of forces and moments

The vertical forces produced by the KUBeetle for various flapping frequencies without changing the

control inputs were first measured using a 6-axis load cell (Nano 17, ATI Industrial Automation, USA). A 2-cell 7.4 V lithium polymer battery can power the KUBeetle to flap its wings at a maximum frequency of 35 Hz. However, high vibration in the measurement system when the wings were operating at a higher flapping frequency than 33 Hz caused errors in the load cell sensor for data recording. Therefore, figure 9 shows that frequencies ranging from 25 to 33 Hz were selected for the measurement. The designed body weight of the KUBeetle is about 20 g. Figure 9 shows that by flapping its wings at a higher rate than 29 Hz, the KUBeetle can produce enough vertical force to stay in the air.

A 6-axis load cell was then used to measure the forces and moments generated to prove the control concept [37]. In the test, the forces and moments generated for pitch, roll and yaw control inputs were measured at a flapping frequency of 25 Hz by adjusting the deflection angles manually from -9° to $+9^\circ$. This frequency was selected to minimize any damage to the motor during the measurement and vibration of the flapping mechanism caused by high voltage application. Figure 10(a) shows the average horizontal forces in the x -direction that tilting the wing roots for pitch, roll and yaw controls generated. Changing the deflection angle for pitch control from -9° to $+9^\circ$ generated horizontal force, which varied linearly from -0.9 to 0.9 g force (gf). However, the change of deflection angles for roll and yaw controls did not significantly affect the horizontal

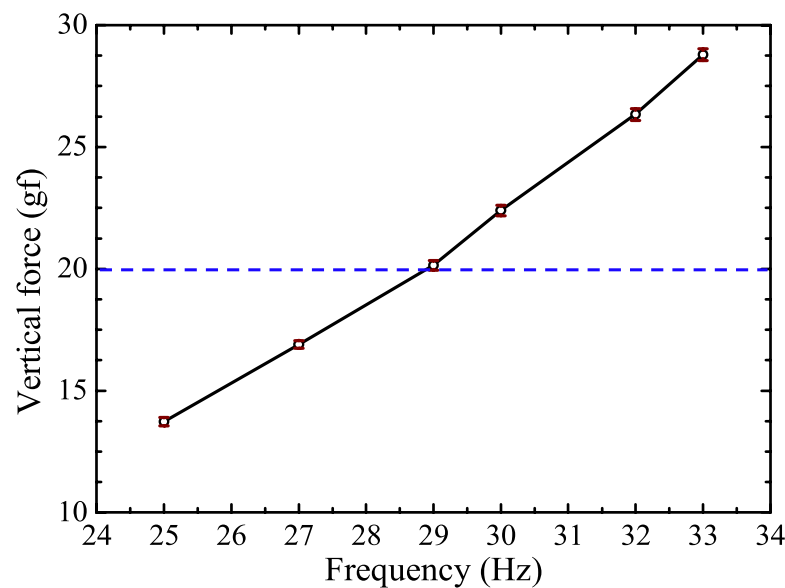


Figure 9. Vertical force generation for various flapping frequencies (1 gf = 9.81 mN, standard deviation (SD) = ± 0.2 gf).

force. Figure 10(b) shows the average lateral forces in the y -direction that change of the control inputs generated. According to the change of control input for roll control from -9° to $+9^\circ$, the lateral force linearly decreased from 1.4 to -0.5 gf. The changes in lateral forces corresponding to control input changes for pitch and yaw were negligible. Figure 10(c) shows that modifying the deflection angles for the three control inputs did not significantly reduce the vertical forces produced in the z -direction. Therefore, the FW-MAV is expected to produce enough vertical force to stay aloft.

Figure 10(d) shows that changing the deflection angle for pitch control from -9° to $+9^\circ$ generated pitch moment that varied from approximately 127 to -124 gf $\cdot$ mm (from 1.246 to -1.216 N $\cdot$ mm). Applying other control inputs for roll and yaw caused insignificant changes in pitch moment. Similarly, figure 10(e) shows that varying only the deflection angle for roll control from -9° to $+9^\circ$ produced roll moment linearly from approximately 160 to -135 gf $\cdot$ mm (from 1.570 to -1.324 N $\cdot$ mm). Figure 10(f) shows that changing the deflection angles for pitch and roll controls resulted in a small change to the yaw moment. However, rotating the wing roots in both wings in the opposite directions produced and varied the yaw moment. Figures 10(d)–(f) show that for the same amount of control input, the magnitude of the yaw moment is smaller than those for pitch and roll moments. This is because the pitch and roll moments are produced by asymmetric vertical forces. Meanwhile, the yaw moment is generated by asymmetric horizontal forces in the x -direction in the two wings, without contribution from vertical forces in the z -direction. Consequently, the yaw moment is less sensitive to the yaw control input. However, the video available in the electronic supplementary materials (supplementary

video 1) (stacks.iop.org/BB/12/036006/mmedia) shows that the FW-MAV performed yaw to change its heading direction with these measured yaw moments. Therefore, tilting the wing roots for attitude change can successfully implement the KUBeetle control principle.

3.4. Conceptual design and fabrication of the control system

We designed simple pitch, roll and yaw control mechanisms actuated by three conventional sub-micro servos that can control the position of the wing roots at the trailing edges in the two wings, as shown in figure 11. Figure 11(a) shows that the arm of servo for pitch control is connected to the wing root connector through a coupler. The wing root connector is linked with one end of each root spar, which is rotated about hinge A on the xz -plane. Figure 11(b) illustrates the motion of the servo arm for roll control. The servo arm is extended and connected to the sliding hinge joined with the wing root connector to rotate the root spars about hinges B1 and B2 on the yz -plane. Thus, when the servo arm for pitch control rotates, the wing root connector can slide in the extended arm of roll servo on the xz -plane without modifying the roll control input. Figure 11(c) shows the mechanism for yaw control. The arm of the yaw servo rotates to tilt the wing root connector around the z -axis. Opposite motions of the wing roots generate yaw moment for the yaw control.

Figure 12 shows the actual assembly of the KUBeetle with the pitch, roll and yaw control mechanisms. Three DM-S0020 sub-micro servos with a response time of 0.08 s/ 60° actuated the control mechanisms, which were made of carbon/epoxy. The original weight of each servo was 2.4 g. Trimming the servo case reduced its weight to 1.9 g. A two-cell 7.4 V Li-poly battery pack powered these servos through a 5 V regulator.

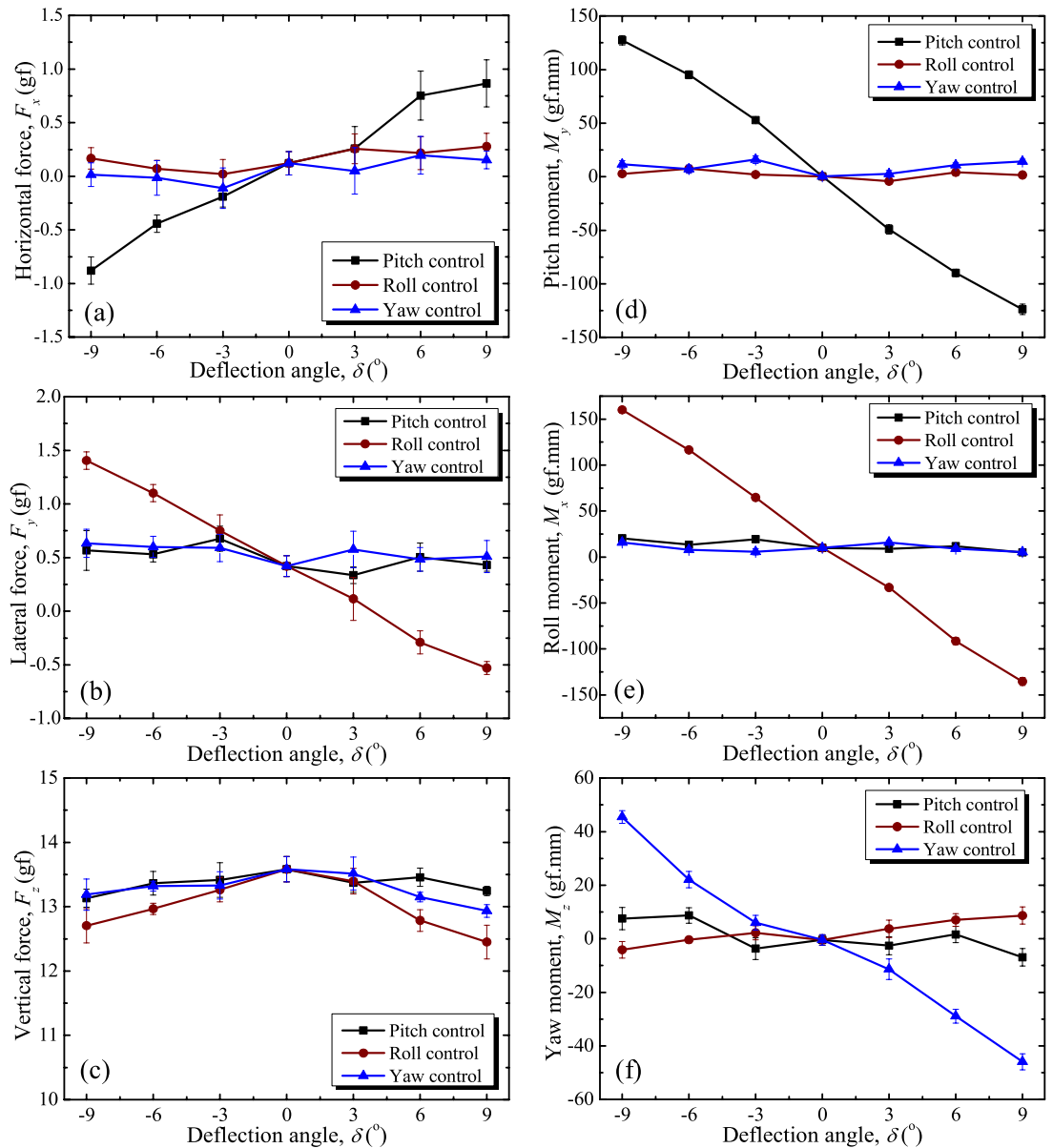


Figure 10. Generation of (a)–(c) forces, and (d)–(f) moments, in three directions, with respect to the variation of the deflection angle, at a flapping frequency of 25 Hz (force SD = ± 0.2 gf, moment SD = ± 3 gf·mm).

4. Implementation of feedback stabilization

4.1. PD control theory and hardware

Without feedback control, the tailless FW-MAV is eventually unstable after takeoff, as shown in figure 13 and in [24, 25, 59, 60]. Therefore, active feedback control should be considered to stabilize the tailless FW-MAV during flight. Among many control methods, a simple PD controller has been successfully installed and demonstrated for stable flight in several FW-MAVs, including the Robobee [25], the Delfly [61, 62], and the Robotic Hummingbird [29], by sensing the three Euler angles. In this work, the PD controller was also implemented to improve the upright stability of the KUBeetle, as figure 14 shows. However, the control outputs are pitch (q), roll (p) and yaw (r) angular rates. Thus, the PD controller installed in the KUBeetle

continuously calculates and minimizes the error between the desired and the measured angular rates by adjusting control variables through the three servos. The PD control algorithm is expressed by the following equation:

$$u(t) = k_P \varepsilon(t) + k_D \frac{d}{dt} \varepsilon(t), \quad (9)$$

where, $u(t)$ is the control input; $\varepsilon(t) = \omega_d(t) - \omega_m(t)$ is the error between the desired rate $\omega_d(t)$ and the measured rate $\omega_m(t)$ from the onboard gyroscope; and k_P and k_D are the proportional and derivative gains, respectively. PD-based in-house code was developed and implemented to a microcontroller to stabilize the KUBeetle during flight. The gains k_P and k_D were experimentally obtained by trial and error.

The hardware for feedback control system installed in the KUBeetle includes a low-power BareDuino

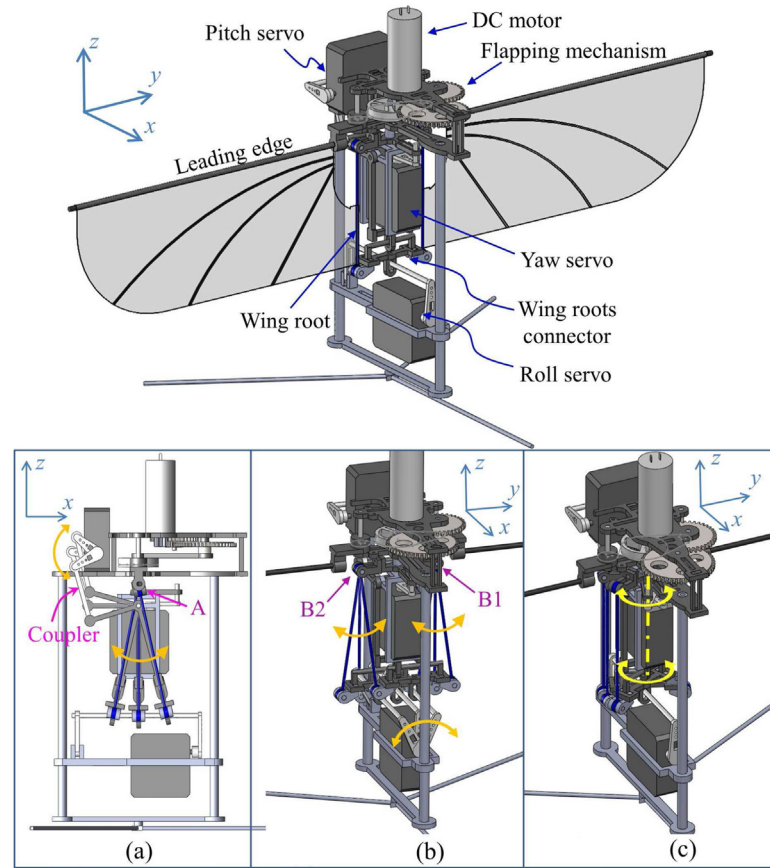


Figure 11. Conceptual design of the mechanism for (a) pitch, (b) roll, and (c) yaw controls.

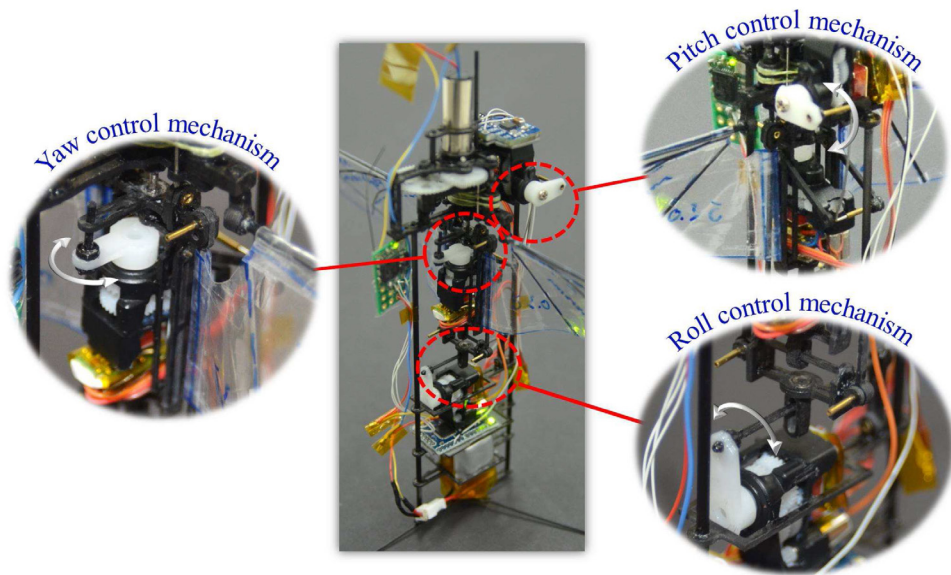


Figure 12. Fabrication of the pitch, roll and yaw control mechanisms. Arrows indicate the motion directions of the servo arms.

Nano microcontroller equipped ATmega328P microprocessor, an L3GD20H 3-axis gyroscope, a Del-tang DT-Rx36 DSM2 compatible receiver, and three Doman DM-S0020 conventional sub-micro servos, as shown in figure 15(a). An onboard gyroscope for stability enhancement was also used in the Nano-Hummingbird [24], Delfly [62], Robobee [63], a 3.2 g FW-MAV [64], and a 13 g ornithopter [65]. A low-pass

filter was used to filter the gyroscope readings $\Omega = [\omega_x \ \omega_y \ \omega_z]^T$ to reduce noise effect, as expressed in the following equation:

$$\omega_t = (1 - \kappa)\omega_{t-1} + \kappa\omega_{\text{gyro},t} \quad (10)$$

where ω_t and ω_{t-1} denotes the filtered angular rates in the current and previous time steps, respectively, κ is the filter coefficient ($\kappa = 0.25$), and $\omega_{\text{gyro},t}$ represents

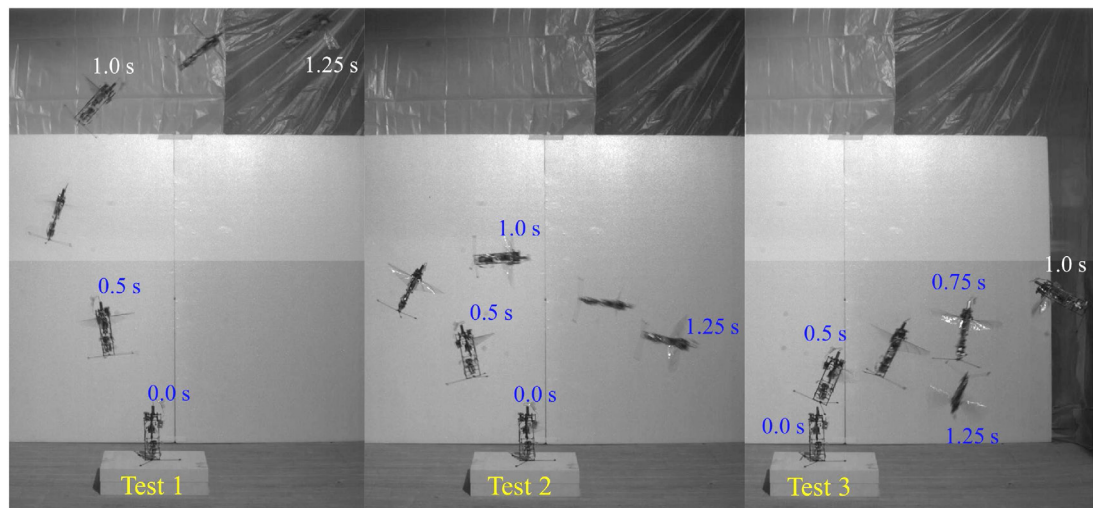


Figure 13. Composite images of the open-loop flights of the KUBeetle captured by a high-speed camera. Videos of the tests are available in the supplementary materials (video 2).

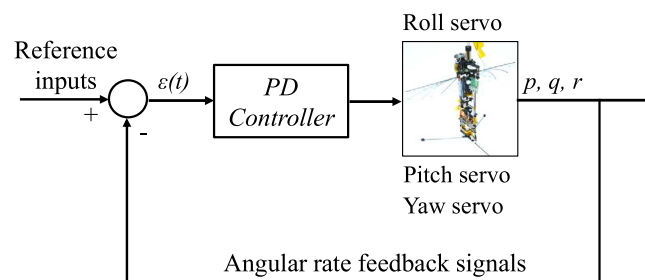


Figure 14. PD feedback control diagram.

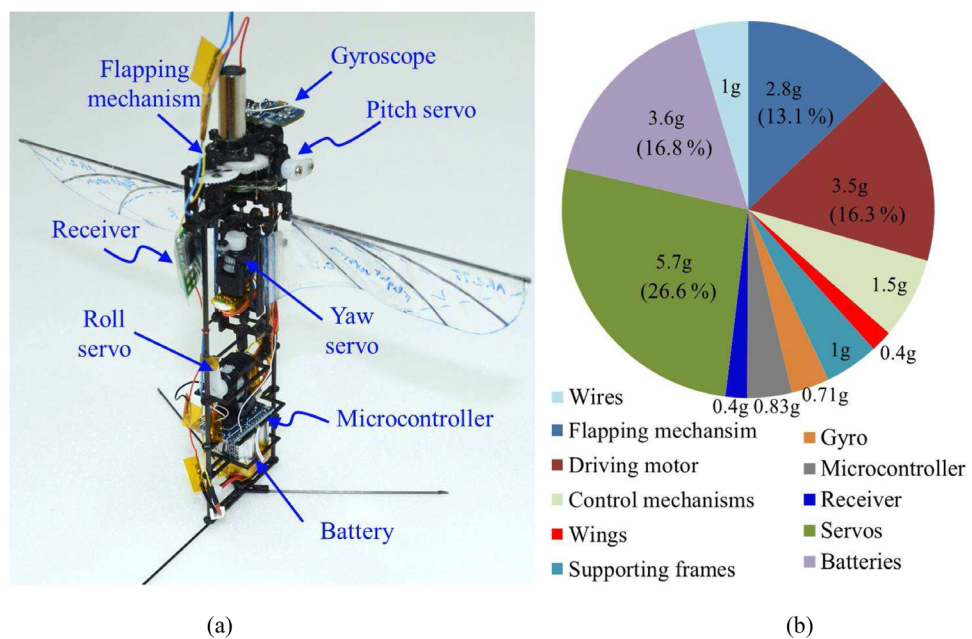


Figure 15. (a) KUBeetle showing the flight control system and (b) the mass distribution of each component.

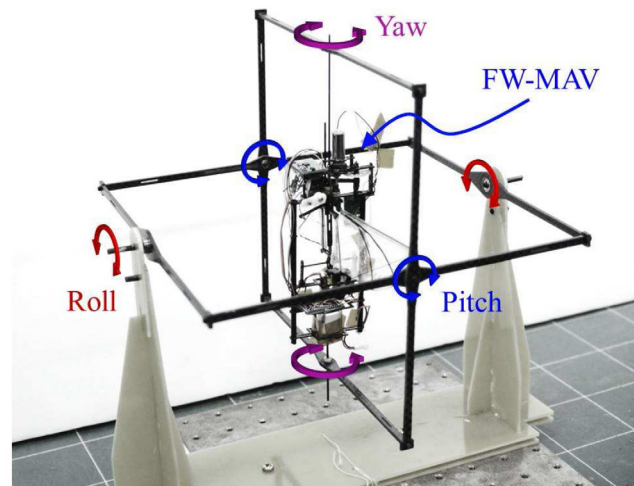


Figure 16. 3DoF gimbal system used for testing controllability and initially obtaining control gains.

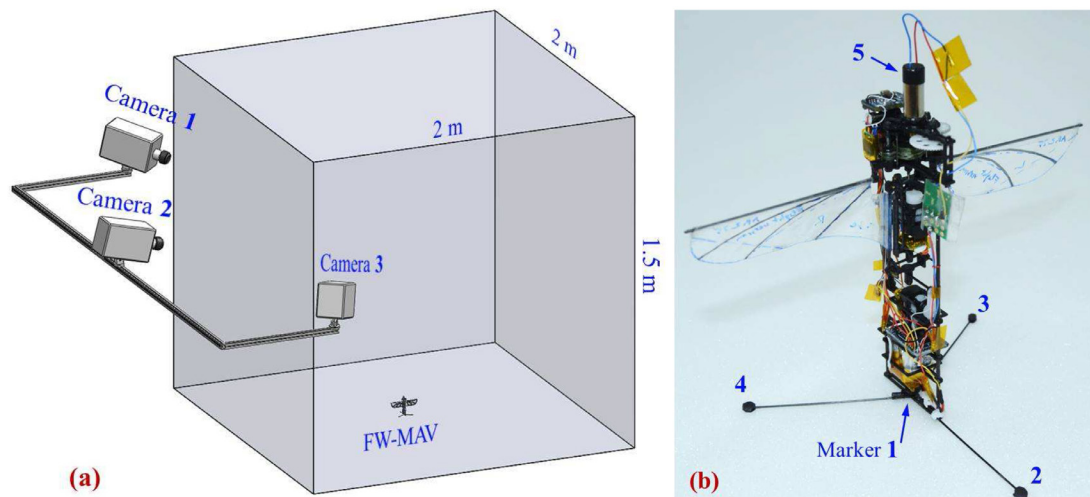


Figure 17. (a) Flight test arena with three synchronized high-speed cameras, and (b) markers on the KUBeetle for motion tracking using the DLT method.

the signals from the gyroscope in the x , y and z axes. Figure 15 shows all of these components installed in the KUBeetle. The total weight of the KUBeetle used for flight was approximately 21.4 g. The weight breakdown is included in figure 15(b). The flapping mechanism, driving motor, battery, and servos account for 72.8% of the total weight.

4.2. Gimbal system

Figure 16 shows that the KUBeetle was installed in a three rotational degrees of freedom (3DoF) gimbal system to investigate the controllability of the proposed control mechanisms. Manufacturing errors may cause unavoidable asymmetric flapping motion between the left and right wings, which may result in pitch and roll motions without any control inputs. To trim these motions, the location of the center of gravity (CG) was adjusted to vertically align with the mean aerodynamic center (AC). This allowed the vehicle to perform a short vertical takeoff without initial moment bias.

Inappropriate control gains applied to the FW-MAV for feedback stability during a flight test may cause damage or failure to the FW-MAV. Therefore, the 3DoF gimbal system was also used to obtain the gains. Due to the added weight of the gimbal system, the control gains obtained to stabilize the FW-MAV in the gimbal system should not be directly used for stable flight. However, the test with the gimbal system was useful as a reference to reduce the number of flight tests to determine the appropriate control gains.

5. Flight tests

5.1. Flight test arena and image recording system

Figure 17(a) shows that the KUBeetle was flight tested in an arena (2 m length $\times$ 2 m width $\times$ 1.5 m height) covered by plastic film to protect and retain the vehicle in the field of view. The KUBeetle was placed at the center of the bottom surface. Three synchronized high-speed cameras (Photron Ultima APX, Japan)

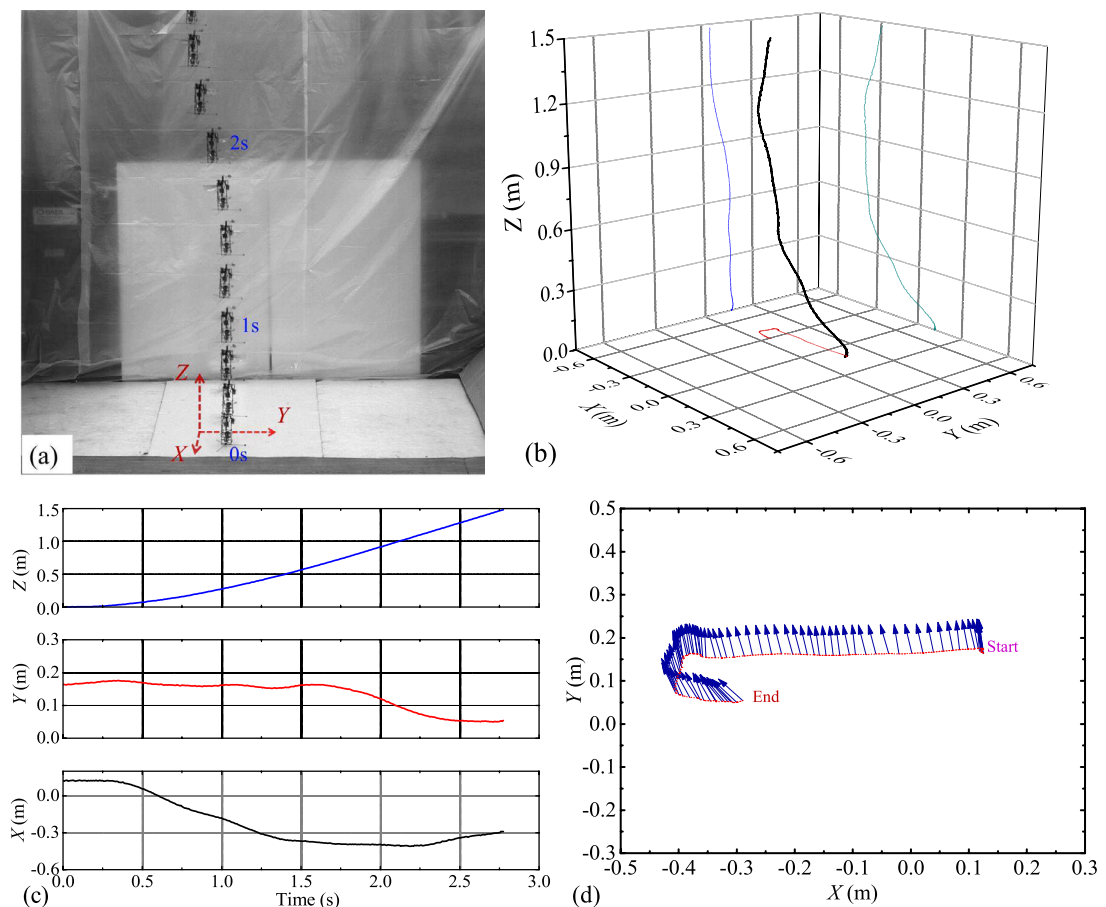


Figure 18. (a) Composite images captured by a high-speed camera, (b) flight trajectory in 3D coordinate, (c) position, and (d) heading direction during vertical takeoff (the arrow direction stands for heading direction).

operating at 125 fps were set up to capture the motion of the vehicle during flight. The DLT method developed by Hedrick [58] was used to track the motion of fixed points located on the vehicle, as shown in figure 17(b).

5.2. Flight results

The KUBeetle was first used to test its stability response during flight by using the gains obtained in the 3DoF gimbal system. Differences in the weight and flapping frequency in the gimbal test and flight test may result in the gains acquired from the gimbal system test being too high or too low for stable flight. Therefore, several flight experiments were conducted to obtain proper control gains.

For flight, a two-cell 7.4 V Li-poly battery pack was used to supply power for the DC motor driver and the microcontroller. The servos and other electronic circuits used a 5 V source, which was regulated from a two-cell 7.4 V battery by an onboard regulator in the microcontroller. A series of flight tests were performed and tracked using three synchronized high-speed cameras filming at 125 fps.

Figure 18 shows that the KUBeetle was first tested for straight vertical takeoff and steady climb out of the field of view. The flapping frequency during the flight was about 30.5 Hz. Figure 18(a) shows composite images of the KUBeetle during vertical takeoff test,

while figures 18(b) and (c) show its flight trajectory and the X, Y, Z positions, respectively. The vehicle required about 3 s to reach an altitude of about 1.5 m. It drifted slightly along the x-axis, but remained upright during flight. The heading direction plotted in figure 18(d) shows the vehicle heading nearly in the same direction during climb up, even though it experienced slight rotation around its body axis (z-axis).

Figure 19 shows the roll, pitch and yaw angles obtained during the first flight test. The small variation in the roll (root-mean-square deviation (RMSD) = 3.260°) and pitch (RMSD = 2.142°) angles from the 0° reference demonstrated that the KUBeetle remained stable and upright during vertical flight. The yaw angle shows a small variation from the 0° reference during the first 2 s, before drifting slightly about the z-axis. Video of the vertical flight test is available in the electronic supplementary materials (supplementary video 3).

Figure 20(a) shows the second flight test, in which the KUBeetle performed a takeoff and hover at a fixed point within the field of view. The flapping frequency in this test was about 30 Hz. Figure 20(b) shows its 3D trajectory, while figure 20(c) shows its position for each coordinate axis. The heading direction represented by the arrow in figure 20(d) showed that the KUBeetle experienced slow rotation about the body axis (z-axis).

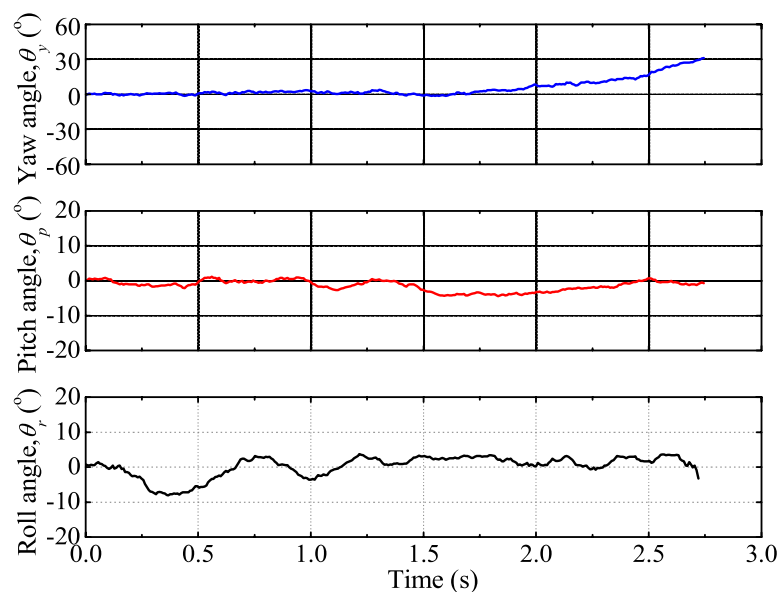


Figure 19. Measured roll, pitch and yaw angles of the KUBeetle during vertical takeoff, obtained by the DLT method.

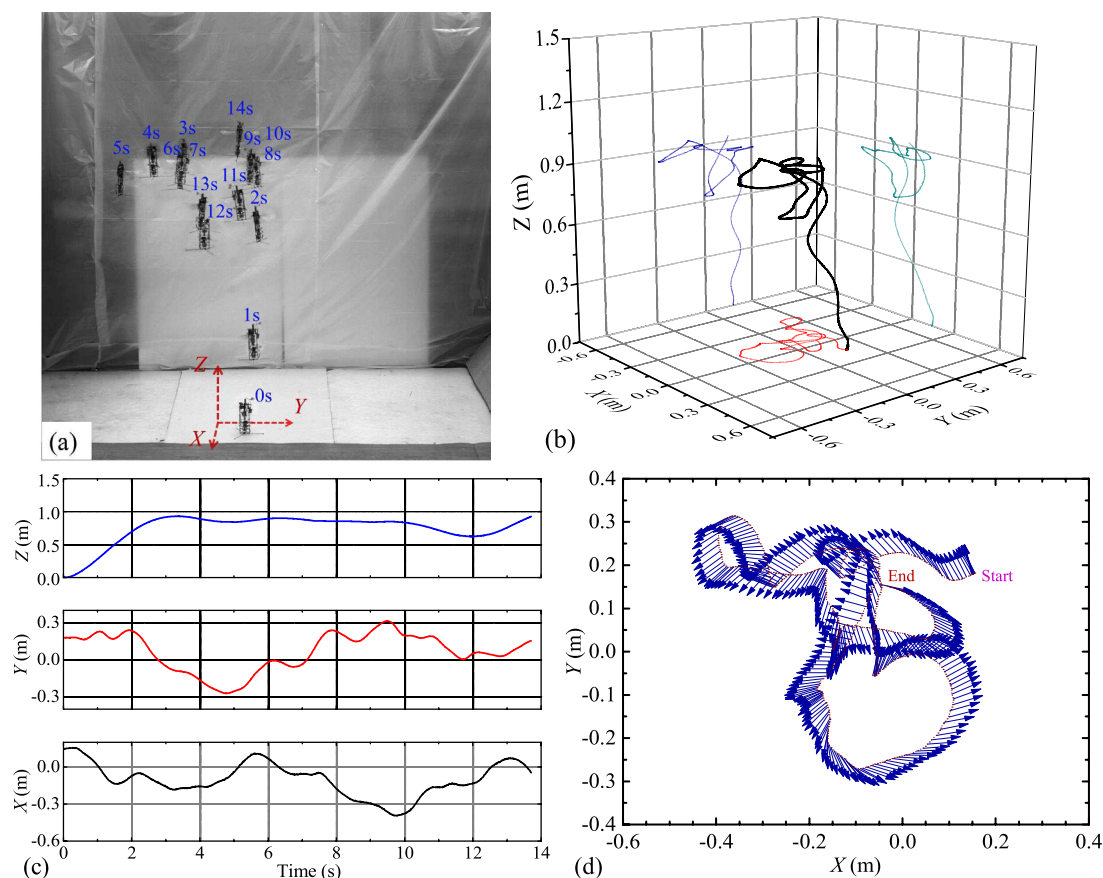


Figure 20. (a) Composite images, (b) flight trajectory in 3D coordinate, (c) position, and (d) heading direction of the KUBeetle during hovering flight (the arrow direction stands for heading direction).

During flight, the vehicle experienced small oscillations in roll ($\text{RMSD} = 3.191^\circ$) and pitch ($\text{RMSD} = 3.412^\circ$) angles from the 0° reference, as shown in figure 21. This might be the result of corrective operation of the feedback control. Figure 20 also shows the yaw angle drift from the initial condition. This is because the initial yaw moment was not perfectly

trimmed and the control gains for yaw were not considerably determined. However, the yawing motion did not significantly affect the upright stability of the vehicle. The second flight test indicated that the vehicle could hover and loiter within a ground radius of 0.3 m. The flight lasted for 14 s. A series of the hovering flight tests were additionally conducted to demonstrate the

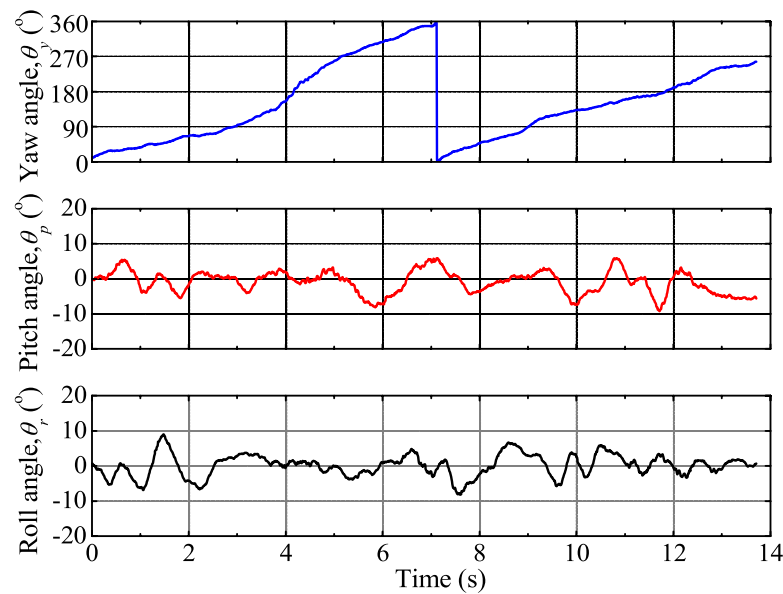


Figure 21. Measured roll, pitch and yaw angles of the KUBeetle during hovering obtained by the DLT method.

stable flight result. Videos of the tests are available in the electronic supplementary materials (supplementary video 4).

5.3. Discussion

Hovering insects, such as locusts, moths, beetles, flies and bumblebees, and tailless flying robots are known to possess inherent instability [24, 25, 66–72]. Without an active feedback system, they are unable to remain stable in flight [73–76]. Indeed, halteres in *Drosophila* [68, 69], ocelli in honeybee [76] and antennae in moths [23, 77] were found to play a role as feedback sensors for achieving flight stability. An experimental study by Ristroph *et al* [74] showed that when their vibratory halteres were disabled by gluing them to their abdomen, fruit flies were unable to maintain stability, and fell down. Sane *et al* [77] found by experiment that without the presence of antennal flagellum, moths could not remain stable in flight. Without feedback control, tailless flying robots also experience a quick tumble [63, 76].

The KUBeetle is eventually unstable after takeoff, as shown in figure 13. Therefore, active feedback is essential to stabilize the vehicle during flight. Among many control strategies, angular rate feedback was shown to provide stability in *Drosophila* flies through their halteres [68, 69, 74, 78], in honeybees through their ocelli [76], and in flying robots through a MEMS gyroscope [24, 63]. Our tailless insect-like KUBeetle could also successfully perform stable vertical flight and hover by using angular rates feedback control alone. However, figure 21 shows that the KUBeetle experienced slight oscillations in pitch and roll motions, which caused the vehicle to loiter around in the capture volume. These oscillations may be caused by slightly imperfect control gains that result in asymmetric control moments. Additionally, the flapping frequency was not under feedback

control. Instead, the flapping frequency was manually controlled through the power throttle stick. Therefore, unstable flapping frequency during flight may affect force generation, thus creating asymmetric corrective moments for stability. Stability of the vehicle can be improved by a further combination of angular rate and attitude feedback control. Future study will address this topic.

6. Conclusion

This work presents the design and stable flight of an insect-like tailless KUBeetle robot that, through a 4-bar linkage and pulley string mechanism, is capable of flapping its wings with high amplitude (190°). The high flapping amplitude means that the KUBeetle is the first two-winged flying robot that successfully uses the clap-and-fling effect at each stroke reversal for force enhancement. In the absence of a tail, the wing rotational modulation mechanism is actuated by three sub-micro servos for KUBeetle attitude control. A series of experimental tests showed that by modifying the rotation angle while flapping its wings, the KUBeetle can produce the desired control moments. A feedback PD controller together with an onboard microprocessor and gyroscope was used to provide pitch, roll and yaw rates for the vehicle. Several flight tests confirmed that our tailless KUBeetle could successfully perform stable and upright flight with very small variations in pitch and roll motions. Further studies that implement feedback for all 6-DoF are needed to improve the flight performance.

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